

Sentiora SRS v1.0

Version 1.0

July 25, 2026

1. Document Information

Field	Value
Product Name	Sentiora
Document	Software Requirements Specification (SRS)
Version	1.0
Status	Initial SRS
Date	July 25, 2026
Based On	Sentiora PRD v2.1 FINAL (July 24, 2026)
Intended Audience	Engineering team, technical architects, QA, design, and future contributors building the Sentiora MVP

This SRS translates the frozen Sentiora PRD v2.1 into implementation-ready requirements. Where the PRD leaves a decision to the Technical Specification, this document preserves that boundary and marks it **TBD — Decision Required** rather than inventing an answer.

2. Introduction

2.1 Purpose

This SRS defines *what the Sentiora system must do* at a level of detail sufficient for a four-member development team to derive architecture, database schemas, APIs, UI screens, GitHub issues, and test cases. It is a translation of product intent (the PRD) into precise, testable system requirements. It does not make architecture, technology, or infrastructure decisions — those belong to the companion Technical Specification.

2.2 Product Scope

Sentiora is a permission-based Digital Life Intelligence Platform. The MVP (v0.1) is a Chrome-only browser extension that, with explicit per-source user consent, captures meaningfully-engaged webpages, PDFs, YouTube videos, and manually saved notes/links; structures this content into searchable personal memory; and allows the user to search that memory and ask natural-language questions answered via Retrieval-Augmented Generation (RAG) with source citations. Footprint (digital identity mapping) and Shield (risk flagging) are explicitly out of scope for the MVP and are addressed only at a product-intent level in this SRS.

2.3 Product Objectives

- O-1: Let users capture meaningful digital information only from sources they explicitly approve.
- O-2: Structure captured information into searchable, attributable personal memory.
- O-3: Provide natural-language search and RAG-based Q&A grounded in the user's own memory, with citations.
- O-4: Keep every act of collection visible, controllable, and reversible.
- O-5: Enforce a permanent, non-overridable exclusion of sensitive content categories.
- O-6: Establish an architecture that can extend to Footprint and Shield without changing the trust model.

2.4 Definitions and Terminology

Term	Definition
Digital Memory / Memory	A structured record capturing one piece of meaningful information, with source, timestamp, excerpt, and metadata.
Digital Source	An origin of content a user can approve for capture (webpage, PDF, YouTube, manual note/link; V1.x: AI conversation, GitHub).
Integration / Connector	The mechanism by which Sentiora connects to and ingests content from a Digital Source.
RAG (Retrieval-Augmented Generation)	A technique where relevant stored content is retrieved and supplied to an LLM before it generates an answer, grounding the response in that content.
Embedding	A vector representation of text used to enable semantic similarity search.
Semantic Search	Search that returns conceptually

	related results based on embedding similarity rather than exact keyword match.
Digital Footprint	(Next phase) The set of accounts, services, and identity surface a user maintains online.
Permission	An explicit, source-scoped user authorization allowing Sentiora to capture data from that source.
Connected Source	A Digital Source for which the user has granted an active permission.
AI Intelligence	The Sentiora Intelligence module's capability to reason over Memory (search, RAG chat, summarization).
Citation	A reference attached to an AI-generated answer that links back to the specific Memory it was derived from.
Data Ingestion	The pipeline step that acquires, processes, and stores approved content as Memory.
OAuth	An authorization protocol used for account-based source connections (applicable to V1.x sources such as GitHub).
Meaningful Capture Engine	The internal filter that decides whether engagement (dwell time, scroll depth, revisits) is sufficient to create a Memory automatically.
Never-Capture / Sensitive-Content Exclusion Policy	The permanent, non-overridable rule set preventing capture from Incognito mode, authentication/password interfaces, payment/financial interfaces, and identifiable private health content.

3. Product Overview

Sentiora is composed of four modules. Memory and Intelligence are in scope for the MVP; Footprint and Shield are next-phase and described here only to preserve architectural continuity.

3.1 Sentiora Memory — “Remember what matters”

- **Purpose:** Capture and structure meaningful information from user-approved sources.
- **User value:** Eliminates digital memory loss; removes the burden of manual organization.
- **Inputs:** Approved webpages, PDFs, YouTube videos, manual notes/links (MVP); AI conversations, GitHub (V1.x).
- **Processing:** Permission check → engagement evaluation (Meaningful Capture Engine) → sensitive-content exclusion check → content extraction → metadata tagging → storage/indexing.
- **Outputs:** Structured Memory records (title, source, timestamp, excerpt, tags) queryable by Intelligence.
- **Dependencies:** None — foundational layer.

3.2 Sentiora Intelligence — “Understand what you know”

- **Purpose:** Reason over Memory to answer questions, search, and summarize.
- **User value:** Removes repeated searching; turns raw memory into usable understanding.
- **Inputs:** Memory index (embeddings + metadata), user natural-language query.
- **Processing:** Query embedding → relevant Memory retrieval → context construction → LLM generation → citation attachment.
- **Outputs:** Grounded, cited answers; ranked search results; topic summaries (V1.x priority for full grounded-summary UX, though semantic search + RAG chat + citations are MVP).
- **Dependencies:** Fully dependent on Memory.

3.3 Sentiora Footprint — “Understand where you exist online” (Next phase)

- **Purpose:** Map accounts, services, auth methods, and identity surface, independent of content browsing.
- **Dependencies:** Own connectors; enriched by Memory where relevant.
- Not defined further in this SRS beyond preserving its module boundary — detailed requirements deferred to the Footprint-phase SRS.

3.4 Sentiora Shield — “Protect your digital identity” (Next phase)

- **Purpose:** Consume Footprint + Memory signals to flag risk and recommend action.
- **Dependencies:** Depends on Footprint (primary) and Memory (secondary); cannot run standalone; performs no independent collection.
- Not defined further in this SRS.

3.5 How the Modules Work Together

Permissions gate Data Sources → Memory ingests and stores approved content → Intelligence retrieves and reasons over Memory to answer user queries. Footprint and Shield are designed to attach to this same permission-first pipeline in later phases without altering the consent model established by Memory.

4. User Types and Roles

Sentiora v0.1 has a single end-user role. No administrative or multi-tenant roles are defined in the PRD, so none are introduced here.

4.1 Registered User

Attribute	Detail
Description	An individual who has created a Sentiora account and uses the Chrome extension and any companion web interface.
Permissions	Full control over their own data: grant/revoke source permissions, initiate capture, search, chat, view, export, delete.
Allowed actions	Create account; connect/disconnect sources; toggle per-source capture; manually save notes/links; browse memory timeline; search (keyword/semantic); ask RAG questions; view citations; delete individual memories; export data (V1.x); delete account (V1.x).
Restricted actions	Cannot access another user's memory, sources, or account data under any condition (strict data isolation — see NFR-SEC series). Cannot enable capture from a permanently excluded sensitive-content category regardless of settings.

No admin, moderator, or support-staff role is defined by the PRD; introducing one is **out of scope** unless a future PRD revision requires it.

5. Functional Requirements

Each requirement below traces to a PRD Functional Requirement ID (Section 12 of the PRD) where one exists, or is marked **Derived Requirement** with justification. Format follows the mandated template.

5.1 Authentication & Account Management

FR-AUTH-001 — Account Creation

- **Requirement Name:** Account creation via email or SSO
- **Description:** The system shall allow a user to create a Sentiora account using email/password or a supported SSO provider.
- **Priority:** Must
- **MVP Status:** MVP
- **Preconditions:** User has not previously registered with the same email/SSO identity.
- **Trigger:** User submits the account-creation form.
- **System Behaviour:** System validates input, creates the account record, and requires verification (email confirmation or SSO token validation) before proceeding.
- **Expected Output:** Verified account is created; user proceeds to onboarding.
- **Failure/Error Behaviour:** Duplicate email, weak password, or failed SSO handshake returns a clear, specific error without exposing internal system state.
- **Acceptance Criteria:** Account is created and verified before any source-connection screen is offered (PRD acceptance criterion, unchanged).
- **Dependencies:** None.
- **Traceability:** PRD FR-AUTH-001.

FR-AUTH-002 — Login (*Derived Requirement*)

- **Description:** The system shall allow a verified user to authenticate with their credentials or SSO to access their account.
- **Priority:** Must
- **MVP Status:** MVP
- **Preconditions:** Account exists and is verified.
- **Trigger:** User submits login credentials.
- **System Behaviour:** Validates credentials; on success, establishes an authenticated session.
- **Expected Output:** User is redirected to the Dashboard.
- **Failure/Error Behaviour:** Invalid credentials return a generic “invalid email or password” message (no indication of which field is wrong, to avoid user enumeration).
- **Acceptance Criteria:** Valid credentials always produce a session; invalid credentials never do.
- **Dependencies:** FR-AUTH-001.
- **Traceability:** Derived Requirement — necessary to operate the account created by FR-AUTH-001; the PRD assumes login exists but does not state it explicitly.

FR-AUTH-003 — Logout (*Derived Requirement*)

- **Description:** The system shall allow a user to end their authenticated session on demand.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Session token is invalidated; extension/UI returns to a logged-out state.
- **Acceptance Criteria:** After logout, previously valid session tokens are rejected.

- **Traceability:** Derived Requirement — standard counterpart to FR-AUTH-002, implied by any authenticated system.

FR-AUTH-004 — Session Management (*Derived Requirement*)

- **Description:** The system shall maintain authenticated sessions with a defined expiry and shall require re-authentication after expiry or logout.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Session tokens expire after a period **TBD** — **Decision Required** (exact duration is a Technical Specification concern).
- **Acceptance Criteria:** Expired sessions are rejected and prompt re-login.
- **Traceability:** Derived Requirement — necessary for NFR-SEC compliance (Section 13.4).

FR-AUTH-005 — OAuth for V1.x Source Connections

- **Description:** The system shall support OAuth-based authentication for connecting account-based V1.x sources (e.g., GitHub) once introduced.
- **Priority:** Should
- **MVP Status:** Post-MVP (V1.x)
- **Traceability:** PRD Section 18 (V1.x — GitHub capture); PRD Section 28 lists “Authentication implementation” as a Technical Specification concern — exact OAuth scopes are **TBD**.

FR-AUTH-006 — Account Deletion

- **Description:** The system shall allow a user to permanently delete their account, which removes all associated memories and connected-source records.
- **Priority:** Should
- **MVP Status:** Post-MVP (V1.x)
- **Preconditions:** User is authenticated and confirms deletion intent explicitly (not a single accidental click).
- **System Behaviour:** Marks account for deletion; propagates deletion across storage, index, and cache within the stated retention window.
- **Expected Output:** Confirmation of deletion is shown/sent to the user.
- **Failure/Error Behaviour:** Partial deletion failures must not leave the account in a silently inconsistent state; retried or flagged for operator follow-up.
- **Acceptance Criteria:** Confirmed account deletion removes all user data within the stated retention window (PRD acceptance criterion, unchanged).
- **Dependencies:** FR-PRIV-005 (individual memory deletion), Section 11 (Deletion propagation, Technical Specification).
- **Traceability:** PRD FR-PRIV-007.

FR-AUTH-007 — Authorization / Access Control (*Derived Requirement*)

- **Description:** The system shall ensure that every request to view, search, modify, or delete memory or source data is scoped strictly to the authenticated user’s own data.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Every data-access operation is checked against the requesting user’s identity; cross-user access attempts are rejected.

- **Acceptance Criteria:** No user can retrieve, search, or act upon another user's memories under any request pattern.
- **Traceability:** Derived Requirement — directly required by PRD Section 15 ("Retrieval is scoped strictly to the asking user's own approved memories") and Privacy by Design (Section 14).

5.2 Permission & Consent Management

FR-UI-001 — Consent-First Onboarding

- **Description:** The system shall present a plain-language explanation of the permission model before any capture toggle is made available.
- **Priority:** Must
- **MVP Status:** MVP
- **Preconditions:** User has created and verified an account.
- **Trigger:** First login / onboarding flow start.
- **System Behaviour:** Onboarding screens explain what is and is not collected; capture toggles remain disabled until the privacy explanation screen has been viewed.
- **Expected Output:** User reaches the source-connection screen with informed understanding.
- **Failure/Error Behaviour:** If onboarding is interrupted, capture remains off by default until onboarding is completed.
- **Acceptance Criteria:** Capture cannot be toggled on until the privacy explanation screen has been viewed (PRD acceptance criterion, unchanged).
- **Dependencies:** FR-AUTH-001.
- **Traceability:** PRD FR-UI-001.

FR-PRIV-001 — Per-Source Toggles (MVP Sources)

- **Description:** The system shall allow the user to independently enable or disable capture for each MVP-supported source: webpages, PDFs, YouTube, and manual notes/links.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Each source has its own independent permission flag; disabling one has no effect on others.
- **Acceptance Criteria:** Disabling one source has no effect on others (PRD acceptance criterion, unchanged). AI conversation and GitHub toggles are excluded from this MVP acceptance test.
- **Traceability:** PRD FR-PRIV-001.

FR-PRIV-001a — Per-Source Toggles (V1.x Sources)

- **Description:** The system shall allow independent enable/disable of capture for AI conversations and GitHub once these sources are introduced.
- **Priority:** Should
- **MVP Status:** Post-MVP (V1.x)
- **Acceptance Criteria:** Deferred to V1.x implementation.
- **Traceability:** PRD FR-PRIV-001a.

FR-PRIV-002 — Sensitive-Content Exclusion Policy

- **Description:** The system shall enforce a permanent, non-overridable exclusion at the point of capture such that no memory is ever created from: Incognito mode; authentication/password-sensitive interfaces; payment/financial-sensitive interfaces; or private health-sensitive content identifiable by supported detection rules.
- **Priority:** Must
- **MVP Status:** MVP
- **Preconditions:** None — applies unconditionally regardless of user

settings.

- **Trigger:** Any capture attempt (automatic or manual-adjacent context evaluation).
- **System Behaviour:** Exclusion check runs before any content is persisted. If a page/context matches an excluded category, capture is blocked and no partial data is written.
- **Expected Output:** No memory record exists for excluded content; user is not shown a false “captured” state for such content.
- **Failure/Error Behaviour:** If exclusion detection is uncertain, the system shall fail closed (do not capture) rather than fail open, for the health-sensitive category specifically, pending detection-rule definition in the Technical Specification.
- **Acceptance Criteria:** No memory is created from Incognito, authentication, or payment/financial interfaces, verified by test pages in each category; health-sensitive exclusion is verified against detection rules defined in the Technical Specification (PRD acceptance criterion, unchanged).
- **Dependencies:** Technical Specification must define exact detection mechanisms, domain rules, classifiers, and false-positive handling — **Technical Feasibility Validation Required.**
- **Traceability:** PRD FR-PRIV-002.

FR-PRIV-003 — Revoke Access

- **Description:** The system shall allow the user to revoke a connected source’s permission at any time.
- **Priority:** Must
- **MVP Status:** MVP
- **Trigger:** User selects “revoke” / “disconnect” for a source.
- **System Behaviour:** Immediately stops future capture from that source. Existing memories previously captured under that permission are **not** automatically deleted.
- **Expected Output:** Source is shown as disconnected; a clear statement that existing memories remain unless separately deleted.
- **Acceptance Criteria:** Capture from that source stops within one session; previously stored memories remain unless separately deleted (PRD acceptance criterion, unchanged).
- **Dependencies:** FR-PRIV-005 (separate deletion action).
- **Traceability:** PRD FR-PRIV-003.

FR-PRIV-004 — Privacy Center

- **Description:** The system shall provide a single screen where the user can view all connected sources and all stored memories, and initiate deletion or export from that screen.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Aggregates connected-source list and memory inventory; every memory is actionable (view/delete) from this screen.
- **Acceptance Criteria:** Every stored memory is visible and actionable from this screen (PRD acceptance criterion, unchanged).
- **Dependencies:** FR-PRIV-005, FR-PRIV-006.
- **Traceability:** PRD FR-PRIV-004.

FR-PRIV-008 — Permission Visibility (*Derived Requirement*)

- **Description:** The system shall clearly display, for each connected source, what data category is permitted to be captured.
- **Priority:** Must
- **MVP Status:** MVP
- **Traceability:** Derived Requirement — required to satisfy PRD Privacy by Design principle “Transparency” (Section 14) which is

stated as a platform-wide principle but not written as a standalone FR in the PRD.

5.3 Digital Source Integrations

For each MVP source, this section defines connection, access boundary, and failure behaviour. Exact API-level integration mechanics are **Technical Feasibility Validation Required** and belong to the Technical Specification.

5.3.1 Webpage

- **Connection process:** Implicit once the “webpage capture” permission toggle (FR-PRIV-001) is enabled in the Chrome extension; no separate account linking required.
- **Authentication mechanism:** None (browser-context based).
- **Required permissions:** Chrome extension host permissions plus the Sentiora “webpage capture” toggle.
- **Data Sentiora may access:** Page content, title, URL, engagement signals (dwell time, scroll depth, revisit count) on pages the user actively browses with the toggle enabled.
- **Data Sentiora must not access:** Incognito-mode pages; pages matching the sensitive-content exclusion policy (FR-PRIV-002); pages visited while the toggle is disabled.
- **Sync behaviour:** Real-time, evaluated per-page-visit against the Meaningful Capture Engine (FR-MEM-001).
- **Failure behaviour:** If content extraction fails, no partial memory is stored; user is notified per FR-UI-002.
- **Disconnect behaviour:** Disabling the toggle stops all future webpage capture immediately (FR-PRIV-003).

5.3.2 PDF

- **Connection process:** Implicit once “PDF capture” toggle is enabled; applies to PDFs opened in-browser.
- **Authentication mechanism:** None.
- **Required permissions:** Extension access to browser-rendered PDF viewer content.
- **Data Sentiora may access:** PDF title, source path/URL, extractable text.
- **Data Sentiora must not access:** PDFs opened in Incognito mode; PDFs matching sensitive-content categories (e.g., a downloaded bank statement) — **Technical Feasibility Validation Required** for reliable detection at the PDF layer.
- **Sync behaviour:** Triggered when a PDF is opened and engagement criteria are met.
- **Failure behaviour:** Unsupported/corrupted PDF fails gracefully with user notification (FR-UI-002); large PDFs are chunked rather than blocking the browsing session.
- **Disconnect behaviour:** Disabling toggle stops future PDF capture.
- **Traceability:** PRD FR-MEM-002.

5.3.3 YouTube

- **Connection process:** Implicit once “YouTube capture” toggle is enabled.
- **Authentication mechanism:** None (public video metadata/transcript access).
- **Required permissions:** Extension access to YouTube page content and available transcript data.
- **Data Sentiora may access:** Video title, metadata, transcript text where available, engagement signals.
- **Data Sentiora must not access:** Videos viewed in Incognito mode.
- **Sync behaviour:** Triggered when engagement criteria are met

during/after video viewing.

- **Failure behaviour:** If no transcript is available, memory is still created with available metadata; this is not treated as a failure.
- **Disconnect behaviour:** Disabling toggle stops future YouTube capture.
- **Traceability:** PRD FR-MEM-003.

5.3.4 Manual Save (Notes/Links)

- **Connection process:** No connection required — always available.
- **Authentication mechanism:** N/A.
- **Required permissions:** None beyond an authenticated session.
- **Data Sentiara may access:** Exactly what the user types or pastes.
- **Sync behaviour:** Immediate, on user action.
- **Failure behaviour:** Manual save must always succeed for well-formed input; if storage fails, the user is notified and the input is not silently lost from the UI (client-side retry/local buffer recommended — **Technical Feasibility Validation Required**).
- **Disconnect behaviour:** N/A — cannot be disabled, as it is independent of passive capture settings.
- **Traceability:** PRD FR-MEM-004.

5.3.5 AI Conversations (V1.x — Post-MVP)

- **Status:** Approved direction, not implemented in MVP. Connection process, auth mechanism, and access boundary to be defined when V1.x work begins.
- **Traceability:** PRD Section 18 (V1.x scope), FR-PRIV-001a.

5.3.6 GitHub (V1.x — Post-MVP)

- **Status:** Approved direction, likely OAuth-based (FR-AUTH-005). Exact scopes and accessible data types are **Technical Feasibility Validation Required**.
 - **Traceability:** PRD Section 18 (V1.x scope), FR-PRIV-001a.
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6. Sentiora Memory Requirements

6.1 Memory Lifecycle

1. **Source connection** — user enables a per-source toggle (Section 5.3).
2. **User-approved data collection** — content is only observed on sources with an active permission.
3. **Data ingestion** — content is passed to the processing pipeline (FR-MEM-001-004).
4. **Processing / normalization** — raw content (HTML, PDF text, transcript) is normalized into a consistent internal representation. Exact representation: **TBD — Technical Specification**.
5. **Metadata extraction** — title, source, timestamp, URL/path, and (where derivable) topic tags are attached.
6. **Categorization** — content is associated with topic tags to support browsing/filtering (FR-MEM-005).
7. **Storage** — normalized content and metadata are persisted; embeddings generated for semantic search (Section 7).
8. **Search / retrieval** — memory becomes queryable via keyword and semantic search (Section 10) and via RAG (Section 7).
9. **Updating** — Memory records are treated as append-only from a content standpoint in MVP; **no requirement exists for editing captured content** (out of scope unless a future PRD revision adds it).
10. **Deletion** — user-initiated, individual memory deletion (FR-PRIV-005); propagates to storage and search index.
11. **Source attribution** — every Memory retains a permanent link to its origin (source type, URL/path, capture timestamp) for citation purposes (Section 16 of PRD).
12. **Duplicate handling** — see FR-MEM-006 below.
13. **Synchronization** — MVP is single-device (Chrome extension); no cross-device sync requirement is defined in the PRD for MVP.

6.2 What Qualifies as a Meaningful Memory

A memory is created automatically only when engagement signals (dwell time, scroll depth, revisits) exceed internally-defined thresholds evaluated by the Meaningful Capture Engine. A page glanced at briefly does **not** become a memory automatically. Manual Save (FR-MEM-004) always succeeds regardless of this evaluation, providing a deterministic fallback.

6.3 Functional Requirements — Memory

FR-MEM-001 — Meaningful Capture Engine

- **Description:** The system shall evaluate engagement signals (dwell time, scroll depth, revisits) against internally-defined thresholds before automatically creating a memory from a webpage.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Below-threshold pages/items do not automatically become memories; Manual Save remains available regardless of the automatic-capture decision.
- **Expected Output:** Memory created only for sufficiently-engaged content.
- **Acceptance Criteria:** Pages/items below the meaningful-engagement threshold do not automatically become memories (PRD acceptance criterion, unchanged). Exact scoring formula and threshold values are a Technical Specification concern, not a user-

facing MVP setting.

- **Traceability:** PRD FR-MEM-001.

FR-MEM-002 — PDF Capture

- **Description:** The system shall allow a user to capture an in-browser PDF as a memory with extracted text.
- **Priority:** Must
- **MVP Status:** MVP
- **Acceptance Criteria:** Captured PDF memory includes title, source path, and extractable text for indexing (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-MEM-002.

FR-MEM-003 — YouTube Capture

- **Description:** The system shall capture a YouTube video as a memory when engagement criteria are met, including title and available transcript.
- **Priority:** Must
- **MVP Status:** MVP
- **Acceptance Criteria:** Memory includes video metadata and transcript text where available (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-MEM-003.

FR-MEM-004 — Manual Save

- **Description:** The system shall let a user manually save a note or link at any time, independent of passive capture settings.
- **Priority:** Must
- **MVP Status:** MVP
- **Acceptance Criteria:** Manual save always succeeds and is timestamped (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-MEM-004.

FR-MEM-005 — Memory Timeline

- **Description:** The system shall let a user browse captured memories chronologically, filterable by source type and date.
- **Priority:** Must
- **MVP Status:** MVP
- **Acceptance Criteria:** Timeline loads and is filterable by source type and date (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-MEM-005.

FR-MEM-006 — Duplicate Handling (*Derived Requirement*)

- **Description:** The system shall detect and deduplicate or link near-identical captures of the same content rather than creating repeated near-identical memories.
- **Priority:** Should
- **MVP Status:** MVP (behaviour required); exact dedup algorithm is Technical Specification concern.
- **Preconditions:** Same source content captured more than once (e.g., a revisited page already meaningfully captured).
- **System Behaviour:** System shall not present the user with visually duplicated memories for the same underlying content.
- **Failure/Error Behaviour:** If dedup cannot be determined confidently, the system may retain both and rely on the user for cleanup rather than silently discarding data — exact tie-breaking is TBD.
- **Acceptance Criteria:** Given two captures of the same URL within a short window, the timeline shows one coherent entry (or clearly linked entries), not unexplained duplicates.

- **Traceability:** PRD Section 20 (Edge Cases — “Duplicate information captured twice”).

6.4 Edge Cases and Failure Conditions (Memory)

Case	Required Behaviour
Unsupported file type	Capture fails gracefully with a clear, non-blocking notification (FR-UI-002).
Very large document	Content is chunked for processing; user browsing is not blocked.
Failed ingestion	User is notified with a retry option; no partial/corrupted memory is silently stored.
Original source later removed online	Captured memory persists; any later Intelligence response using it must note the live source may be unreachable.
Empty memory state	Search/chat clearly state there is nothing to search yet and prompt the user to connect a source.

7. Sentiora Intelligence Requirements

7.1 RAG Workflow

User Query → Query Processing → Relevant Memory Retrieval → Context Construction → LLM Processing → Response → Source Citations

Each stage operates strictly within the scope of the asking user's own approved memories (FR-AUTH-007).

7.2 Functional Requirements — Intelligence

FR-INT-001 — RAG Chat

- **Description:** The system shall let a user ask natural-language questions and receive answers grounded in retrieved memory.
- **Priority:** Must
- **MVP Status:** MVP
- **Preconditions:** At least one memory exists (otherwise see FR-INT-004).
- **System Behaviour:** Query is embedded, relevant memories retrieved, context constructed, and passed to the LLM to generate a grounded response.
- **Expected Output:** A response where every substantive claim is traceable to at least one retrieved memory.
- **Failure/Error Behaviour:** If the LLM provider is unavailable, the user receives a clear error rather than a silently degraded or fabricated answer.
- **Acceptance Criteria:** Every substantive claim in a response is traceable to at least one retrieved memory (PRD acceptance criterion, unchanged).
- **Dependencies:** FR-SEARCH-001, FR-INT-002.
- **Traceability:** PRD FR-INT-001.

FR-INT-002 — Source Citations

- **Description:** The system shall display citations linking back to originating memories on every RAG response.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Citation includes source name/type and capture date; clicking opens the source memory and the relevant excerpt.
- **Acceptance Criteria:** Clicking a citation opens the source memory (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-INT-002; PRD Section 16 (Trust & Explainability).

FR-INT-003 — Grounded Summaries

- **Description:** The system shall let a user request a summary of memories on a topic, citing the memories it draws from and excluding unsupported claims.
- **Priority:** Should
- **MVP Status:** Post-MVP (V1.x) — natural extension of RAG chat, not required for the MVP's essential loop (Semantic Search + RAG Chat + Source Citations).
- **Acceptance Criteria:** Summary cites the memories it draws from and does not include unsupported claims (PRD acceptance criterion, unchanged).

- **Traceability:** PRD FR-INT-003.

FR-INT-004 — Insufficient Evidence Handling (Derived Requirement)

- **Description:** The system shall explicitly state that it does not have enough information rather than generating an unsupported answer when retrieval yields insufficient relevant memory.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** If retrieval confidence/relevance falls below a workable threshold, response explicitly communicates insufficient evidence instead of guessing.
- **Acceptance Criteria:** No response presents an unsupported claim as fact drawn from the user's memory.
- **Traceability:** PRD Section 15 ("Behavior in Boundary Conditions" table) — this is a direct restatement of a PRD-mandated behaviour that was not given its own FR ID in the PRD.

FR-INT-005 — Conflicting Memory Handling (Derived Requirement)

- **Description:** When retrieved memories conflict, the system shall surface both/all conflicting memories with their sources rather than silently resolving the conflict.
- **Priority:** Must
- **MVP Status:** MVP
- **Traceability:** PRD Section 15 (Boundary Conditions) and Section 20 (Edge Cases — "Conflicting information across memories").

FR-INT-006 — Unavailable Source Handling (Derived Requirement)

- **Description:** When a cited memory's original source is no longer reachable, the system shall use the cached memory content and note in the response that the original source may no longer be reachable.
- **Priority:** Should
- **MVP Status:** MVP
- **Traceability:** PRD Section 15 (Boundary Conditions), Section 20 (Edge Cases).

7.3 AI Behaviour Boundary Table

Condition	Required Behaviour
Relevant information exists	Answer generated with citations to supporting memories.
Information insufficient	System states it doesn't have enough information (FR-INT-004).
Multiple conflicting memories	Both surfaced with sources (FR-INT-005).
Source unavailable	Cached content used; response notes possible unreachability (FR-INT-006).
Model uncertain	Uncertainty communicated explicitly, not presented as confident fact.

The system shall never fabricate a claim and attribute it to the user's memory. This is a hard product constraint, not a best-effort behaviour.

8. Sentiora Footprint Requirements (*Next Phase — Not in MVP*)

The PRD explicitly defers Footprint’s detailed scope to a future Footprint-phase specification (PRD Decisions Register, Section 27-A: “Footprint MVP scope: Intentionally deferred”). Per PRD Section 5, Footprint will:

- Map accounts, services, and identity surface through explicit connectors the user approves.
- Reflect only what users explicitly connect or disclose — it will not claim to discover every account or piece of personal data on the internet.
- Be enriched by, but not dependent on, Memory content.

Data Sentiora actually has permission to access: account/service metadata from explicitly connected Footprint connectors (once built).

Data Sentiora may infer or summarize: references to services mentioned within already-approved Memory content (e.g., a service name appearing in a captured page), clearly distinguished from directly-connected Footprint data.

No FR-FOOT requirement beyond the placeholder below is defined until the Footprint-phase PRD/SRS exists.

FR-FOOT-001 — Footprint Dashboard (*Placeholder*)

- **Description:** User can view mapped accounts and services.
 - **Priority:** N/A (next phase)
 - **MVP Status:** Post-MVP — Next Phase
 - **Acceptance Criteria:** To be defined in the Footprint-phase specification.
 - **Traceability:** PRD FR-FOOT-001.
-

9. Sentiora Security

Requirements (*Product-Facing; Next Phase for Shield*)

This section covers product-facing security/awareness functionality (Shield), distinct from backend security controls (Section 13.4). Per the PRD, Shield is entirely next-phase and depends on Footprint, so no MVP FR-SEC (product) requirements are defined here beyond noting the constraint below.

- Shield shall not independently collect data; it consumes only Footprint and Memory signals (PRD Section 5.4).
- Shield shall issue recommendations, not automated actions, unless the user explicitly and separately authorizes a specific action (PRD Decisions Register, Section 27-A: “Additional consent tiers”).
- Sentiora shall not imply detection of information that connected APIs/connectors do not actually expose.

No further Security(Shield) functional requirements are defined until the Shield-phase specification exists.

10. Search Requirements

FR-SEARCH-001 — Semantic Search

- **Description:** The system shall let a user search memory using natural language, returning conceptually related results even without exact keyword overlap.
- **Priority:** Must
- **MVP Status:** MVP
- **System Behaviour:** Query is embedded and compared against memory embeddings via similarity search; results ranked by relevance and recency.
- **Acceptance Criteria:** A semantically related query returns relevant memories even without exact keyword overlap (PRD acceptance criterion, unchanged).
- **Traceability:** PRD FR-SEARCH-001.

FR-SEARCH-002 — Keyword Search (*Derived Requirement*)

- **Description:** The system shall support keyword search returning memories containing matching terms, ranked by relevance and recency.
- **Priority:** Should
- **MVP Status:** MVP (as a complement to semantic search)
- **Traceability:** PRD Section 15 (“Search & Intelligence Experience” table lists Keyword search as an expected mode) — not given its own FR ID in the PRD but explicitly specified as required behaviour.

FR-SEARCH-003 — Filtering and Sorting (*Derived Requirement*)

- **Description:** The system shall allow filtering search/timeline results by source type and date, and sorting by relevance or recency.
- **Priority:** Should
- **MVP Status:** MVP
- **Traceability:** Derived from FR-MEM-005’s “filterable by source type and date” acceptance criterion, extended to search results for consistency.

FR-SEARCH-004 — Empty and Failed Search Handling (*Derived Requirement*)

- **Description:** The system shall clearly communicate when a search returns no results, and shall surface a non-blocking error if the search operation itself fails.
- **Priority:** Should
- **MVP Status:** MVP
- **Traceability:** PRD Section 20 (Edge Cases — “Empty memory (no sources connected yet)”), extended to the general no-results case.

10.1 Search vs. AI Question Answering

Search (keyword/semantic, FR-SEARCH-001/002) returns a ranked list of matching Memory records for the user to browse themselves. RAG Chat (FR-INT-001) goes further: it retrieves relevant memories *and* generates a synthesized, cited natural-language answer. Search never fabricates content; RAG synthesizes but must remain strictly grounded (Section 7.3).

11. Data Management Requirements

ID	Requirement	Priority	MVP Status
FR-DATA-001 (Derived)	The system shall create a Memory record only after the permission and sensitive-content checks pass.	Must	MVP
FR-DATA-002 (Derived)	The system shall store Memory content and metadata with a permanent link to source attribution.	Must	MVP
FR-DATA-003 (Derived)	The system shall allow retrieval of any of the user's own Memory records via timeline, search, or citation click-through.	Must	MVP
FR-PRIV-005	The system shall allow a user to permanently delete a single memory; it is removed from storage and the search index within the session.	Must	MVP
FR-PRIV-006	The system shall allow a user to export their memory data (content and metadata, all sources) in a portable format.	Should	Post-MVP (V1.x)
FR-DATA-004 (Derived)	The system shall enforce strict data isolation such that no user's memory is retrievable by another user under any query.	Must	MVP
FR-DATA-005 (Derived)	The system shall deduplicate/synchronize repeated captures per FR-MEM-006.	Should	MVP

Note: Memory records are **not user-editable** in MVP (no update/edit requirement exists in the PRD) — content updates are limited to deletion and recapture. Data ownership rests entirely with the user (PRD Section 26, Data Control Model); Sentiora acts as custodian, not owner.

12. Privacy Requirements

Privacy is a cross-cutting constraint (PRD Section 14). The following requirements formalize it.

ID	Requirement	Priority	MVP Status
FR-PRIV-009 (Derived)	The system shall require explicit, affirmative consent before accessing any source — no pre-checked boxes, no consent implied by inaction.	Must	MVP
FR-PRIV-010 (Derived)	The system shall collect only the minimum information required for the selected functionality (data minimization), enforced via the Meaningful Capture Engine and per-source toggles.	Must	MVP
FR-PRIV-002	The system shall never capture from Incognito mode, authentication/password interfaces, payment/financial interfaces, or identifiable private health content, regardless of user settings.	Must	MVP
FR-PRIV-003	The system shall allow revocation of any source's permission at any time, stopping future capture immediately.	Must	MVP
FR-PRIV-005	The system shall allow deletion of individual memories at any time.	Must	MVP
FR-PRIV-006	The system shall allow export of memory data.	Should	Post-MVP
FR-PRIV-011 (Derived)	AI processing (RAG, embeddings) shall operate only on already-approved Memory content and shall never process or store content from unapproved sources.	Must	MVP
FR-PRIV-012 (Derived)	Operational/system logs shall never contain passwords, raw authentication tokens, API secrets, or other credentials (see Section 18).	Must	MVP

What Sentiora must never do without user permission: capture from any source lacking an active per-source toggle; capture from any permanently-excluded sensitive-content category under any circumstance; share, sell, or expose one user's memory to another user or unauthorized third party; use captured content for any purpose beyond providing Memory/Intelligence functionality to that same user.

13. Non-Functional Requirements

13.1 Performance

ID	Requirement	Target (Proposed)
NFR-PERF-001	Dashboard/timeline page load	Proposed: under 2 seconds for typical memory volumes at MVP scale.
NFR-PERF-002	Search response time	Proposed: a few seconds under typical MVP load (PRD-stated target, not a contractual SLA).
NFR-PERF-003	RAG chat response time	Proposed: a few seconds under typical MVP load (PRD-stated target, not a contractual SLA).
NFR-PERF-004	Background capture processing	Shall not block active browsing; processing occurs asynchronously.

Exact numeric targets beyond the PRD’s “a few seconds” guidance are **TBD** — to be established from Alpha/Beta baseline testing per PRD Section 19.

13.2 Scalability

ID	Requirement
NFR-SCALE-001	Storage and indexing shall scale per-user without requiring architectural rework before the Footprint/Shield phases.
NFR-SCALE-002	The system shall be designed for MVP-scale usage (early-adopter cohort: students, researchers, developers) rather than enterprise-scale concurrency; unrealistic enterprise assumptions are explicitly avoided.

13.3 Availability & Reliability

ID	Requirement
NFR-AVAIL-001	Capture failures shall degrade gracefully (skip and notify) rather than blocking browsing.
NFR-AVAIL-002	Integration/source failures (e.g., token expiry, unreachable page) shall not silently corrupt Memory data.
NFR-AVAIL-003	AI provider failures shall surface a clear error to the user rather than a fabricated or silently degraded response.
NFR-AVAIL-004	Database failures shall not result in partial/corrupted Memory writes; failed writes shall be retried or clearly reported.

13.4 Security

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ID	Requirement
NFR-SEC-001	All data in transit shall be encrypted.
NFR-SEC-002	Access to stored memory shall require an authenticated session (see FR-AUTH-002, FR-AUTH-007).
NFR-SEC-003	Passwords shall never be stored in plaintext (exact hashing scheme: TBD — Technical Specification).
NFR-SEC-004	OAuth tokens and other credentials shall be stored securely and never logged in plaintext (see FR-PRIV-012).
NFR-SEC-005	The system shall apply rate limiting to authentication and API endpoints to reduce abuse risk. Exact thresholds: TBD .
NFR-SEC-006	All user-supplied input shall be validated/sanitized before processing or storage.
NFR-SEC-007	Secrets (API keys, provider credentials) shall be managed via a secure secret-management mechanism, not hardcoded or committed to source control.
NFR-SEC-008	Session tokens shall be protected against common theft/replay vectors (exact mechanism: TBD — Technical Specification).
NFR-SEC-009	Data at rest shall be encrypted where applicable (PRD-stated target — exact scope of “where applicable” is a Technical Specification decision).
NFR-SEC-010	Cross-user access control shall be enforced at every data-access layer (see FR-AUTH-007, FR-DATA-004).

13.5 Privacy

ID	Requirement
NFR-PRIV-001	No memory shall be created outside explicitly approved sources.
NFR-PRIV-002	The never-capture list (FR-PRIV-002) shall be enforced at the point of capture, not after the fact.

13.6 Usability

ID	Requirement
NFR-USE-001	Core navigation (Dashboard, Timeline, Search, Chat, Privacy Center) shall be understandable without external documentation.
NFR-USE-002	Permission screens shall use plain language, avoiding legal/technical jargon.
NFR-USE-003	AI citations shall be visually clear and directly clickable to their source.
NFR-USE-004	Delete and disconnect controls shall be easy to find and require an unambiguous confirmation step for destructive actions.
NFR-USE-005	The UI shall be responsive across standard desktop browser window

	sizes (Chrome extension + any companion web UI).
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13.7 Accessibility

ID	Requirement
NFR-ACC-001	Core screens (timeline, search, chat, Privacy Center) shall support keyboard navigation.
NFR-ACC-002	Core screens shall meet baseline contrast ratio guidance (WCAG 2.1 AA as a reasonable MVP target — not explicitly mandated by the PRD, noted as Assumption ASM-006).
NFR-ACC-003	Interactive elements shall carry screen-reader-accessible labels.

13.8 Maintainability

ID	Requirement
NFR-MAINT-001	Module boundaries (Memory / Intelligence / Footprint / Shield) shall remain independently deployable as the product grows.
NFR-MAINT-002	Capture, ingestion, and deletion events shall be logged for debugging and for user-facing activity transparency (see Section 18).
NFR-MAINT-003	Configuration (e.g., capture thresholds, provider endpoints) shall be externalized rather than hardcoded, to support tuning without redeployment.
NFR-MAINT-004	Core ingestion, retrieval, and permission-enforcement logic shall be covered by automated tests given the four-person team's maintenance constraints.

13.9 Compatibility

ID	Requirement
NFR-COMPAT-001	The MVP client shall support the Chrome browser only (current stable Chrome release; exact minimum version TBD).
NFR-COMPAT-002	No mobile app or non-Chrome browser support is required for MVP (PRD Non-Goals, Section 4).

14. External Interface Requirements

14.1 User Interface

Per PRD Section 17, the following screens are required. Detailed UI mockups are explicitly out of scope for this SRS, consistent with the PRD.

Screen	Purpose	Major Actions	MVP Status
Landing / Signup / Login	Entry point; account creation and authentication.	Create account, log in, initiate SSO.	MVP
Onboarding	Explain the permission model before capture is enabled.	View privacy explanation, proceed to source connection.	MVP
Dashboard	Home screen; entry point for daily use.	View recent memories, quick search, launch chat.	MVP
Connected Sources	Manage active source permissions.	Enable/disable per-source toggles, view connection status.	MVP
Memory Timeline	Chronological browse of captured memories.	Filter by source/date, open a memory, delete a memory.	MVP
Ask Sentiora (Search / AI Chat)	Natural-language search and RAG chat.	Enter query, view results/answer, follow citations.	MVP
Topics	Group memories by topic.	Browse by topic tag.	MVP
Privacy Center	Central control point for data.	View all sources/memories, delete, export (V1.x).	MVP
Digital Footprint	View mapped accounts/services.	<i>(Next phase — not built in MVP.)</i>	Next Phase
Account Settings	Manage account-level settings.	Update credentials, initiate account deletion (V1.x).	MVP (basic) / V1.x (deletion)

14.2 API Interfaces

The following API categories are anticipated; exact endpoint design belongs to the Technical Specification (PRD Section 28).

- **Authentication APIs** — account creation, login, logout, session validation.
- **User APIs** — profile/account settings.
- **Permission APIs** — grant/revoke/view per-source permissions.
- **Integration APIs** — source connection handshake (webpage/PDF/YouTube capture triggers; OAuth for V1.x sources).
- **Memory APIs** — create (ingest), read (timeline, single memory), delete memory.
- **Search APIs** — keyword and semantic search.
- **AI APIs** — RAG query submission, response with citations, summary generation (V1.x).

- **Footprint APIs** (*Next phase*) — account/service discovery.
- **Security APIs** (*Next phase*) — Shield risk signals.

14.3 External Services

Service Category	Purpose	Data Exchanged	Auth Method	Dependency Risk	Privacy Considerations
Embedding model provider	Generate vector embeddings for semantic search/RAG.	Memory text content (already user-approved).	Provider API key (server-side).	Medium — provider change affects re-indexing; selection is TBD.	Content must be approved before use; to already approved Memory only (FI 011).
LLM provider (RAG generation)	Generate grounded chat responses.	Retrieved memory context + user query.	Provider API key (server-side).	Medium-High — outage triggers NFR-AVAIL-003.	Must not be used to generate content; the query user's content approved Memory
Chrome Web Store / Extension platform	Distribution of the MVP client.	Extension package, permission manifest.	Developer account.	Low-Medium — policy changes could affect required permissions.	Extension permissions request match data access; describe onboarding
YouTube (page-level access)	Source for video capture.	Video metadata, transcript (where available).	None (browser-context).	Medium — page structure changes could break capture; flagged Technical Feasibility Validation Required.	Only can be used with appropriate permissions; engage threshold
SSO provider(s) (if used)	Alternative account authentication.	Identity token / verified email.	OAuth.	Low-Medium.	No passwords stored; account

15. Data Requirements

Major entities implied by the functional requirements above. Detailed schema design belongs to a separate Database Design Document.

Entity	Purpose	Key Information	Relationships
User	Represents a registered account.	Email/identity, auth credentials or SSO reference, account status, created date.	Owns Permissions, ConnectedSources, Memories, AIConversations.
Permission	Represents a per-source consent grant.	Source type, status (active/revoked), granted date, revoked date.	Belongs to User; governs ConnectedSource behaviour.
ConnectedSource	Represents an active source connection.	Source type, connection metadata (e.g., OAuth reference for V1.x).	Belongs to User; governed by Permission.
Memory	A single structured capture.	Title, source type, source URL/path, timestamp, excerpt/content, topic tags, embedding reference.	Belongs to User; originates from a ConnectedSource; referenced by Citation.
MemorySource (Derived)	Attribution linkage between a Memory and its originating capture event.	Capture timestamp, engagement signal snapshot (for audit/debugging).	Belongs to Memory.
AIConversation	A RAG chat thread.	Created date, associated User.	Belongs to User; contains AIMessages.
AIMessage	A single chat turn.	Query or response text, timestamp, role (user/assistant).	Belongs to AIConversation; assistant messages reference Citations.
Citation	Links an AI response claim to a source Memory.	Referenced Memory ID, excerpt used.	Belongs to AIMessage; references Memory.
FootprintInsight (Next phase — placeholder only)	Would represent a mapped account/service.	Not defined in MVP.	Not defined in MVP.
SecurityInsight (Next phase — placeholder only)	Would represent a Shield-generated risk flag.	Not defined in MVP.	Not defined in MVP.

16. System Workflows

Workflow 1 — User Registration & Onboarding

- **Actor:** Prospective user.
- **Preconditions:** None.
- **Steps:** Visit signup → provide email/SSO → verify account → view consent-first onboarding (FR-UI-001) → reach source-connection screen.
- **System Actions:** Create account (FR-AUTH-001); gate capture toggles until onboarding viewed.
- **Result:** Verified, onboarded user ready to connect sources.
- **Failure Cases:** Duplicate email; verification failure; onboarding abandoned (capture remains off).

Workflow 2 — Connecting a Digital Source

- **Actor:** Registered user.
- **Preconditions:** Onboarding completed.
- **Steps:** Open Connected Sources screen → select a source → review requested access → enable toggle (FR-PRIV-001).
- **System Actions:** Create/activate Permission record; enable capture pipeline for that source.
- **Result:** Source shown as connected; capture becomes active.
- **Failure Cases:** OAuth handshake failure (V1.x sources); permission denied at browser level.

Workflow 3 — Granting Permissions

- **Actor:** Registered user.
- **Preconditions:** Source selected.
- **Steps:** View plain-language description of requested access → confirm.
- **System Actions:** Persist consent record with timestamp.
- **Result:** Source-specific capture is authorized.
- **Failure Cases:** User declines — no permission is created, no capture occurs.

Workflow 4 — Data Ingestion into Sentiora Memory

- **Actor:** System (triggered by user browsing behaviour or manual save).
- **Preconditions:** Active permission for the relevant source.
- **Steps:** Content observed → sensitive-content exclusion check (FR-PRIV-002) → engagement evaluation (FR-MEM-001, if automatic) → extraction and normalization → metadata tagging → storage and indexing.
- **System Actions:** Create Memory record; generate embedding.
- **Result:** New searchable Memory available.
- **Failure Cases:** Exclusion match (no memory created); below-threshold engagement (no automatic memory; manual save still available); extraction failure (FR-UI-002 notification, no partial memory).

Workflow 5 — Searching Memories

- **Actor:** Registered user.
- **Preconditions:** At least one Memory exists (otherwise empty-state messaging).

- **Steps:** Enter query → system performs keyword and/or semantic search → results ranked and returned.
- **System Actions:** Query embedding (semantic path); ranking by relevance/recency.
- **Result:** Ranked list of matching memories.
- **Failure Cases:** No matches (empty-state message); search backend failure (FR-SEARCH-004).

Workflow 6 — Asking Sentiora Intelligence a Question

- **Actor:** Registered user.
- **Preconditions:** At least one relevant Memory ideally exists (system still responds honestly if not).
- **Steps:** Enter natural-language question → relevant memory retrieval → context construction → LLM generates grounded answer → citations attached and displayed.
- **System Actions:** Per RAG workflow (Section 7.1).
- **Result:** Cited, grounded answer or an honest “insufficient information” response.
- **Failure Cases:** LLM provider outage (NFR-AVAIL-003); no relevant memory (FR-INT-004); conflicting memories (FR-INT-005).

Workflow 7 — Viewing Digital Footprint (Next Phase)

- Not implemented in MVP; placeholder only, per PRD Section 5 and Decisions Register.

Workflow 8 — Viewing Security Insights (Next Phase)

- Not implemented in MVP; placeholder only, per PRD Section 5 and Decisions Register.

Workflow 9 — Revoking Source Permission

- **Actor:** Registered user.
- **Preconditions:** An active connected source exists.
- **Steps:** Open Connected Sources or Privacy Center → select source → revoke.
- **System Actions:** Deactivate Permission; immediately stop future capture for that source; existing Memory records untouched.
- **Result:** Source shown as disconnected; historical memories remain, clearly labeled as no longer actively syncing.
- **Failure Cases:** Revocation attempted on an already-inactive source (idempotent no-op).

Workflow 10 — Deleting User Data / Account

- **Actor:** Registered user.
- **Preconditions:** User is authenticated.
- **Steps (individual memory):** Open Privacy Center or Timeline → select a memory → confirm delete (FR-PRIV-005).
- **Steps (account):** Open Account Settings → initiate account deletion → confirm explicitly (FR-AUTH-006).
- **System Actions:** Propagate deletion across storage, search index, and cache.
- **Result:** Deleted memory no longer appears in timeline, search, or future RAG retrieval; past chat answer text may remain as history, but re-asking no longer retrieves the deleted memory.
- **Failure Cases:** Partial deletion failure — must not leave system in

a silently inconsistent state (NFR-AVAIL-002).

17. Error Handling Requirements

Scenario	Required Behaviour
Invalid login	Generic, non-enumerable error message (Section 5.1, FR-AUTH-002).
Expired session	User is prompted to re-authenticate; request is rejected until then.
Expired OAuth token (V1.x sources)	User is notified and prompted to reconnect; no silent failure (PRD Section 20).
Revoked third-party permission	Treated as a disconnect; future capture stops; existing memories retained.
Integration/API failure	Capture attempt fails gracefully with notification (FR-UI-002); no partial memory.
Network failure	Client-side retry or clear offline indication; no silent data loss for manual save.
Database failure	No partial/corrupted writes (NFR-AVAIL-004); operation reported as failed to the user where relevant.
AI provider failure	Clear error shown; no fabricated response (NFR-AVAIL-003).
Search failure	Non-blocking error message (FR-SEARCH-004).
Data ingestion failure	Notification with retry option (FR-UI-002); no silently stored partial memory.
Duplicate data	Deduplicated or linked per FR-MEM-006.
Unauthorized request	Rejected at the authorization layer (FR-AUTH-007); generic error, no internal detail exposed.
Rate limit exceeded	Clear, user-understandable message; no internal system detail exposed.

All user-facing errors shall avoid exposing internal system state, stack traces, or implementation details (see NFR-SEC series).

18. Logging & Monitoring Requirements

Shall be logged: - Application errors (with enough context for debugging, excluding sensitive content). - Integration/source failures (capture attempt failures, token expiry). - Authentication events (login success/failure, logout, session expiry) — without credentials. - Data processing/ingestion failures. - AI provider failures (timeouts, errors). - Security-relevant events (e.g., repeated failed logins, access-control rejections).

Shall NOT be logged, under any circumstance: - Passwords (hashed or plaintext). - Raw authentication tokens (OAuth access/refresh tokens, session tokens). - API secrets / provider keys. - Other credentials.

Distinction: Operational/system logs (above) are strictly separate from the user's personal Memory content. Memory content shall never be written into general-purpose operational logs; any content needed for debugging an ingestion failure shall be handled through a mechanism that respects the same access controls as Memory itself — exact mechanism **TBD — Technical Specification**.

19. Security Threat Considerations

Threat	System Requirement / Mitigation Expectation
Unauthorized access	Authenticated sessions required for all data access (NFR-SEC-002); authorization checks on every request (FR-AUTH-007).
Broken access control	Strict per-user data isolation enforced at every access layer (FR-DATA-004, NFR-SEC-010).
Token theft	Secure token storage and handling; tokens never logged (NFR-SEC-004, NFR-SEC-008).
Data leakage	Encryption in transit and at rest where applicable (NFR-SEC-001, NFR-SEC-009); minimized data collection (FR-PRIV-010).
Cross-user data exposure	Isolation enforced at query level; no shared indexes without per-user scoping (FR-DATA-004).
Prompt injection	RAG context construction shall treat retrieved memory content as data, not instructions, to the LLM; exact prompt-hardening approach is TBD — Technical Specification , but the requirement that untrusted captured content must not be able to override system instructions is a hard constraint.
Malicious content from connected sources	Captured content is treated as untrusted input through the ingestion pipeline; sanitized before storage/rendering (NFR-SEC-006).
API abuse	Rate limiting on authentication and API endpoints (NFR-SEC-005).
Injection attacks	Input validation/sanitization on all user-supplied input (NFR-SEC-006).
Account takeover	Secure credential storage (NFR-SEC-003), session protections (NFR-SEC-008), rate-limited login attempts (NFR-SEC-005).
Sensitive data exposure	Permanent sensitive-content exclusion policy (FR-PRIV-002) prevents ingestion of the highest-risk categories at the source.

A full threat model is out of scope for this SRS and belongs to the Technical Specification / a dedicated security review.

20. Constraints

Confirmed Constraints (from PRD)

- MVP client is Chrome-only; no other browsers or mobile apps (PRD Non-Goals, Section 4).
- Development team size is four members — MVP scope is intentionally narrow to fit this capacity (PRD Section 18).
- No always-on activity logging is permitted under any configuration (PRD Non-Goals).
- The sensitive-content exclusion policy is permanent and non-overridable — no product configuration may weaken it (PRD Section 27-A).
- AI responses must never fabricate claims presented as coming from user memory (PRD Section 15).

Assumed Constraints

- **ASM-CON-001:** The team will rely on third-party embedding and LLM provider APIs rather than training/hosting custom models in MVP, consistent with a four-person team's realistic capacity. *(Assumption — not stated explicitly in the PRD, which defers "Embedding model and LLM/model provider" to the Technical Specification.)*
 - **ASM-CON-002:** Legal/regulatory compliance (e.g., GDPR/CCPA-style obligations) has not been formally assessed, per the PRD's own Risk table (Section 21: "Legal / compliance considerations... flagged as an open item for legal review"). This SRS does not assert compliance with any specific regulatory framework.
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21. Assumptions & Dependencies

Confirmed Dependencies (explicit in PRD)

- **DEP-001:** Intelligence depends entirely on Memory; it has nothing to reason over without it (PRD Section 5).
- **DEP-002:** Shield depends on Footprint (primary) and Memory (secondary); cannot function standalone (PRD Section 5).
- **DEP-003:** Footprint can use its own connectors but is enriched by the same permission/source layer as Memory (PRD Section 25).
- **DEP-004:** RAG chat (FR-INT-001) depends on Semantic Search (FR-SEARCH-001) and Source Citations (FR-INT-002) to satisfy the MVP's essential-loop definition.

Assumptions

- **ASM-001:** Users must explicitly connect external accounts/sources before Sentiora can ingest their data. (*Directly restates PRD's own example assumption.*)
 - **ASM-002:** A companion web application (beyond the Chrome extension) is needed to host full-screen experiences (Dashboard, Privacy Center, Timeline) — the PRD describes these screens without specifying whether they live inside the extension popup or a separate web app. **Decision Required** in the Technical Specification.
 - **ASM-003:** Single-device usage is acceptable for MVP; no cross-device memory sync is required, since the PRD does not mention multi-device sync for v0.1.
 - **ASM-004:** Session/token expiry duration, password hashing scheme, and rate-limit thresholds are engineering decisions with no PRD-specified values; reasonable industry-standard defaults will be chosen in the Technical Specification.
 - **ASM-005:** Memory content is not user-editable post-capture in MVP; only deletion is supported, since the PRD defines no edit requirement.
 - **ASM-006:** WCAG 2.1 AA is a reasonable baseline accessibility target, since the PRD states only “baseline accessibility practices” without naming a standard.
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22. MVP Scope

Requirement/Feature	Module	Priority	MVP?	Dependency
Account creation (FR-AUTH-001)	Auth	Must	Yes	—
Login/Logout/Session (FR-AUTH-002-004)	Auth	Must	Yes	FR-AUTH-001
Authorization/data isolation (FR-AUTH-007)	Auth	Must	Yes	FR-AUTH-002
Consent-first onboarding (FR-UI-001)	Memory/Privacy	Must	Yes	FR-AUTH-001
Per-source toggles, MVP sources (FR-PRIV-001)	Memory/Privacy	Must	Yes	FR-UI-001
Sensitive-content exclusion policy (FR-PRIV-002)	Memory/Privacy	Must	Yes	—
Revoke access (FR-PRIV-003)	Memory/Privacy	Must	Yes	FR-PRIV-001
Privacy Center (FR-PRIV-004)	Memory/Privacy	Must	Yes	FR-PRIV-005, 006
Delete individual memory (FR-PRIV-005)	Memory	Must	Yes	—
Meaningful Capture Engine (FR-MEM-001)	Memory	Must	Yes	FR-PRIV-001
PDF capture (FR-MEM-002)	Memory	Must	Yes	FR-PRIV-001
YouTube capture (FR-MEM-003)	Memory	Must	Yes	FR-PRIV-001
Manual save (FR-MEM-004)	Memory	Must	Yes	—
Memory timeline (FR-MEM-005)	Memory	Must	Yes	Memory records exist
Duplicate handling (FR-MEM-006)	Memory	Should	Yes	FR-MEM-001-004
Semantic search (FR-SEARCH-001)	Intelligence	Must	Yes	Memory records + embeddings
Keyword search, filter/sort (FR-SEARCH-002/003)	Intelligence	Should	Yes	Memory records
Empty/failed search handling (FR-SEARCH-004)	Intelligence	Should	Yes	—
RAG chat (FR-INT-001)	Intelligence	Must	Yes	FR-SEARCH-001
Source citations (FR-INT-002)	Intelligence	Must	Yes	FR-INT-001
Insufficient-evidence handling (FR-INT-004)	Intelligence	Must	Yes	FR-INT-001
Conflicting-memory handling (FR-INT-005)	Intelligence	Must	Yes	FR-INT-001
Unavailable-source handling (FR-INT-006)	Intelligence	Should	Yes	FR-INT-001
Ingestion failure feedback (FR-UI-002)	Memory	Should	Yes	—
Grounded summaries (FR-INT-003)	Intelligence	Should	No — V1.x	FR-INT-001/002
Data export (FR-PRIV-	Privacy	Should	No —	Memory data

006)			V1.x	
Account deletion (FR-AUTH-006/FR-PRIV-007)	Auth/Privacy	Should	No — V1.x	FR-PRIV-005
AI conversation capture (FR-PRIV-001a, source)	Memory	Should	No — V1.x	New connector
GitHub capture (FR-PRIV-001a, source, FR-AUTH-005)	Memory	Should	No — V1.x	OAuth, new connector
Additional browser support	Platform	Should	No — V1.x	—
Sentiora Footprint (all)	Footprint	Could	No — Next phase	New module
Sentiora Shield (all)	Shield	Could	No — Next phase	Footprint live
Topic clustering / proactive recommendations	Intelligence	Could	No — V2+	Mature memory corpus

No PRD-defined MVP feature has been moved to Post-MVP in this table; only PRD-designated V1.x/Next-phase items are marked as such. Feasibility concerns (e.g., health-content detection reliability) are flagged separately in Sections 5.2 and 20 rather than used to silently descope an MVP feature.

23. Out of Scope

Explicitly out of scope for the Sentiora MVP, per the PRD:

- Sentiora Footprint (digital footprint mapping) — next phase.
- Sentiora Shield (risk flagging, protective recommendations) — next phase.
- Browsers other than Chrome.
- Mobile applications.
- Automated account-level protective actions (e.g., auto-revoking a third-party permission on Sentiora's own initiative).
- Any form of always-on activity logging.
- Data export and account deletion (deferred to V1.x, though designed for from the start).
- AI conversation and GitHub source capture (deferred to V1.x).
- Editing captured Memory content (no such requirement exists in the PRD).
- Multi-user/team/admin roles (no such requirement exists in the PRD).

Sentiora is explicitly not a hidden surveillance or indiscriminate activity-tracking system. It does not capture all browsing activity; it captures only meaningfully-engaged content from sources the user has explicitly and individually approved, and it permanently excludes an entire class of sensitive content regardless of configuration.

24. Acceptance Criteria

High-level, testable acceptance criteria for the complete Sentiora MVP:

- **AC-001:** A user can connect a supported digital source only after granting the required permission for that specific source.
 - **AC-002:** A user can revoke a connected source's access, and future capture from it stops within the same session.
 - **AC-003:** No memory is ever created from Incognito mode, authentication/password interfaces, or payment/financial interfaces, verified by dedicated test pages for each category.
 - **AC-004:** A user can manually save a note or link successfully at any time, regardless of automatic capture settings.
 - **AC-005:** A webpage below the meaningful-engagement threshold does not automatically become a memory, while Manual Save for the same page still succeeds.
 - **AC-006:** A user can browse a chronological memory timeline, filterable by source type and date.
 - **AC-007:** A semantically related search query returns relevant memories even without exact keyword overlap.
 - **AC-008:** Every RAG chat response's substantive claims are traceable to at least one cited memory, and clicking a citation opens the source memory.
 - **AC-009:** When no relevant memory exists for a question, the system states it doesn't have enough information rather than fabricating an answer.
 - **AC-010:** A user can delete an individual memory, and it no longer appears in timeline, search, or future RAG retrieval.
 - **AC-011:** The Privacy Center displays every connected source and every stored memory, each actionable from that single screen.
 - **AC-012:** A failed capture/ingestion attempt surfaces a visible, non-blocking notification with a retry option, and never silently stores a partial or corrupted memory.
 - **AC-013:** No user can retrieve or act upon another user's memories under any tested request pattern.
-

25. Requirements Traceability Matrix

Req ID	Requirement	PRD Section	Module	Priority	MVP/Post-MVP
FR-AUTH-001	Account creation	12	Auth	Must	MVP
FR-AUTH-002	Login	Derived	Auth	Must	MVP
FR-AUTH-003	Logout	Derived	Auth	Must	MVP
FR-AUTH-004	Session management	Derived	Auth	Must	MVP
FR-AUTH-005	OAuth for V1.x sources	18, 28	Auth	Should	Post-MVP
FR-AUTH-006	Account deletion	12 (FR-PRIV-007)	Auth/Privacy	Should	Post-MVP
FR-AUTH-007	Authorization / access control	15, 14	Auth	Must	MVP
FR-UI-001	Consent-first onboarding	12	Memory	Must	MVP
FR-PRIV-001	Per-source toggles (MVP)	12	Memory	Must	MVP
FR-PRIV-001a	Per-source toggles (V1.x)	12	Memory	Should	Post-MVP
FR-PRIV-002	Sensitive-content exclusion	12, 14	Memory	Must	MVP
FR-PRIV-003	Revoke access	12, 11	Memory/Privacy	Must	MVP
FR-PRIV-004	Privacy Center	12	Privacy	Must	MVP
FR-PRIV-005	Delete individual memory	12	Memory/Privacy	Must	MVP
FR-PRIV-006	Data export	12	Privacy	Should	Post-MVP
FR-PRIV-008	Permission visibility	Derived (Sec. 14)	Privacy	Must	MVP
FR-MEM-001	Meaningful Capture Engine	12	Memory	Must	MVP
FR-MEM-002	PDF capture	12	Memory	Must	MVP
FR-MEM-003	YouTube capture	12	Memory	Must	MVP
FR-MEM-004	Manual save	12	Memory	Must	MVP
FR-MEM-005	Memory timeline	12	Memory	Must	MVP
FR-MEM-006	Duplicate handling	Derived (Sec.	Memory	Should	MVP

		20)			
FR-SEARCH-001	Semantic search	12	Intelligence	Must	MVP
FR-SEARCH-002	Keyword search	Derived (Sec. 15)	Intelligence	Should	MVP
FR-SEARCH-003	Filtering/sorting	Derived	Intelligence	Should	MVP
FR-SEARCH-004	Empty/failed search	Derived (Sec. 20)	Intelligence	Should	MVP
FR-INT-001	RAG chat	12	Intelligence	Must	MVP
FR-INT-002	Source citations	12, 16	Intelligence	Must	MVP
FR-INT-003	Grounded summaries	12	Intelligence	Should	Post-MVP
FR-INT-004	Insufficient evidence	Derived (Sec. 15)	Intelligence	Must	MVP
FR-INT-005	Conflicting memory handling	Derived (Sec. 15, 20)	Intelligence	Must	MVP
FR-INT-006	Unavailable source handling	Derived (Sec. 15, 20)	Intelligence	Should	MVP
FR-UI-002	Ingestion failure feedback	12	Memory	Should	MVP
FR-FOOT-001	Footprint dashboard	12	Footprint	N/A	Next Phase

Derived Requirement legend: Requirements marked “Derived” were not given an explicit ID in the PRD but are necessary consequences of an explicitly stated PRD behaviour, principle, or acceptance criterion; the PRD source is cited in each row.

26. Open Questions / Decisions Required

ID	Question	Why It Matters	Recommended Decision	Blocking?
Q-001	Where do full-screen experiences (Dashboard, Timeline, Privacy Center) live — inside the extension UI, or a separate companion web app?	Affects architecture, auth flow, and hosting.	Recommend a lightweight companion web app authenticated via the same account system, with the extension handling capture + quick actions.	Blocking
Q-002	What are the exact engagement thresholds (dwell time, scroll depth, revisit count) for the Meaningful Capture Engine?	Directly determines capture quality and noise; explicitly deferred by the PRD.	To be tuned empirically during Alpha; start with conservative defaults and iterate from user feedback.	Blocking (for FR-MEM-001 implementation)
Q-003	What are the exact detection rules for the health-sensitive content exclusion category?	Directly affects privacy guarantees (FR-PRIV-002); PRD explicitly defers this.	Recommend starting with a conservative domain/keyword allowlist-of-exclusion approach and fail-closed behaviour, refined pre-launch.	Blocking
Q-004	Which embedding model and LLM provider will be used?	Affects cost, latency, quality, and data-handling agreements.	Deferred to Technical Specification; PRD explicitly leaves this open.	Blocking
Q-005	What is the exact session/token expiry duration and password hashing scheme?	Security-critical; no PRD-specified value.	Recommend industry-standard defaults (e.g., short-lived access token + refresh token; bcrypt/argon2 hashing) — to be confirmed in Technical Specification.	Non-blocking (defaults can be applied and revised)
Q-006	How is deduplication (FR-MEM-006) determined technically — exact URL matching, content-hash similarity, or semantic similarity?	Affects both storage growth and answer quality.	Recommend starting with normalized-URL + content-hash matching, layering semantic similarity later if needed.	Non-blocking

Q-007	What retention window applies to account deletion (FR-AUTH-006)?	User-facing commitment; PRD states “documented deletion/retention process” without a number.	To be defined and published before V1.x account deletion ships.	Non-blocking for MVP (feature is V1.x)
Q-008	What is Sentiora’s stance on GDPR/CCPA-style compliance?	Legal exposure; PRD explicitly flags this as unassessed.	Recommend formal legal review before any broader/public launch beyond the early-adopter cohort.	Blocking (pre-public-launch, not pre-MVP-build)
Q-009	What prompt-hardening approach prevents captured content from being interpreted as instructions by the LLM (prompt injection risk)?	Security-critical given untrusted third-party content is ingested.	Recommend strict role/content separation in prompt construction, treating retrieved memory strictly as data; finalize in Technical Specification.	Blocking

End of Sentiora Software Requirements Specification v1.0. Based on Sentiora PRD v2.1 FINAL. Architecture, schema, and API design belong to the companion Technical Specification.